Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/07/2025
PI Name:	Narayanan, Anand Ph.D.		
Project Title:	Partial Gravity and Sex-Difference Effects on the Venous Circulation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Immunology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	anarayan09@gmail.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203 785 6055
Organization Name:	Florida State University		
PI Address 1:	College of Human Sciences		
PI Address 2:	242 Sandels Building, 120 Convocation Way		
PI Web Page:			
City:	Tallahassee	**State:**	FL
Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Former email address: anarayanan@fsu.edu		
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	06/01/2023	**End Date:**	05/31/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2025 per NSSC information (Ed., 5/22/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Delp, Michael Ph.D. (Florida State University)		
Grant/Contract No.:	80NSSC23K0885		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Space Biology aims to understand organism adaptations to the spaceflight environment. An understudied area of space biology are the spaceflight adaptations of the venous circulation, a component of the cardiovascular system that also includes the heart, arteries, and lymphatics. The venous circulation supports the cardiovascular system, and, in turn, all organs of the body, by transporting blood and its contents away from organ systems toward the heart and lungs. Consequently, venous return impacts cardiac function. However, our knowledge of venous adaptations to spaceflight is minimal, and furthermore, may play a relevant role with astronaut medical conditions such as Spaceflight Associated Neuro-ocular Syndrome (SANS). Recently, it was shown venous return of blood from the head was impaired in astronauts while in space, resulting in the formation of venous thromboemboli (AMA Netw Open. 2019). We also have preliminary evidence in mice from Rodent Research-9 suggesting jugular vein functional and molecular adaptations after spaceflight exposure. As NASA aims to send astronauts to the Moon and Mars, understanding the adaptations of the venous circulation to microgravity, lunar gravity, and Martian gravity will not only increase our knowledge of cardiovascular function in space along a gravity continuum, but also identifying areas of potential medical risks in astronauts. Furthermore, NASA will be sending the first female astronaut into deep space as part of the Artemis Moon venture, and to date there exists no literature on possible sex differences of the venous circulation in response to spaceflight. I hypothesize the venous circulation show impaired function, structural, and molecular changes across simulated Martian, lunar-, and microgravity exposure, increasing in the level of dysfunction as the gravitational loading decreases. Furthermore, the female venous circulation will show a greater effect to the gravity continuum versus male venous circulation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In completing these studies, I have shown how the vascular state of the lymphatic circulation from the head as it relates with Spaceflight Associated Neuro-ocular Syndrome adapts to various spaceflight conditions (actual spaceflight, simulated microgravity and/or radiation, etc.). These studies will enhance our understanding of space biological adaptations of the cardiovascular system, and furthermore, increase our understanding of medical conditions astronauts develop while in space. In studying spaceflight biomedical adaptations, we also increase our knowledge of Earth-based biomedical science by investigating novel pathways activated by the unique spaceflight environment (i.e., gravity, low-dose radiation, isolation, etc.) that may provide new perspectives of how medical conditions develop and/or biology evolve on Earth. These studies provide foundational work that I will further characterize through various ongoing and future projects.
Task Progress:	NASA Space Biology aims to understand organism adaptations to the spaceflight environment. An understudied area of space biology are the spaceflight adaptations of the venous circulation, a component of the cardiovascular system that also includes the heart, arteries, and lymphatics. The venous circulation supports the cardiovascular system, and, in turn, all organs of the body, by transporting blood and its contents away from organ systems toward the heart and lungs. Consequently, venous return impacts cardiac function. However, our knowledge of venous adaptations to spaceflight is minimal, and furthermore, may play a relevant role with astronaut medical conditions such as Spaceflight Associated Neuro-ocular Syndrome (SANS). Recently, it was shown venous return of blood from the head was impaired in astronauts while in space, resulting in the formation of venous thromboemboli (JAMA Netw Open, 2019). We also have preliminary evidence in mice from NASA's Rodent Research-9 mission (Ed. Note: see https://), suggesting jugular vein functional and molecular adaptations after spaceflight exposure. As NASA aims to send astronauts to the Moon and Mars, understanding the adaptations of the venous circulation to microgravity, lunar gravity, and Martian gravity will not only increase our knowledge of cardiovascular function in space along a gravity continuum, but also identify areas of potential medical risks in astronauts. Furthermore, NASA will be sending the first female astronaut into deep space as part of the Artemis Moon venture, and to date there exists no literature on possible sex differences of the venous circulation in response to spaceflight. I hypothesize the venous circulation show impaired function, structural, and molecular changes across simulated Martian, lunar-, and microgravity exposure, increasing in the level of dysfunction as the gravitational loading decreases. Furthermore, the female venous circulation will show a greater effect to the gravity continuum versus male venous circulation. Given project requirements, I have been required to adapt the original proposal to the following research and aims, focusing on studies of the lymphatic vasculature instead, a complementary system to the veins, and studying their physiological adaptations in similar conditions as was originally proposed (e.g. long-term effects of simulated microgravity and/or radiation exposure). Thus, the new proposed aims include: Aim 1: To determine whether simulated microgravity and/or deep space radiation has long-term effects on the physiology of cervical lymphatics involved with spaceflight-associated neuro-ocular syndrome. OCTOBER 2025 UPDATE Per the Principal Investigator (Ed., 10/9/25): Aim 1 has been completed, specifically the experimental efforts, with statistical analysis being completed and manuscripts in preparation. Preliminary findings with respect to Aim 1 show that, in the context of cardiovascular – specifically, lymphatic physiological function adaptations to simulated microgravity and/or deep space radiation exposure – at certain pressures, the average diastolic diameter for the radiation group was larger compared to the average diameter of the control group. This difference was significant. The measures highlight adaptations of the physiological state of the cervical lymphatics in response to simulated microgravity and/or deep space radiation exposure, associated with fluid drainage in the context of astronaut vision changes and SANS. Ongoing statistical analyses are being completed to finalize the data results and share research outcomes at conferences and publications. REFERENCES Marshall-Goebel K, Laurie SS, Alferova IV, Arbeille P, Auñón-Chancellor SM, Ebert DJ, Lee SM, Macias BR, Martin DS, Pattarini JM, Ploutz-Snyder R. Assessment of jugular venous blood flow stasis and thrombosis during spaceflight. JAMA Netw Open. 2019 Nov 1;2(11):e1915011-. PMID: 31722025 ; http://dx.doi.org/
Bibliography Type:	Description: (Last Updated: 05/22/2024)